23.

Venn Diagram

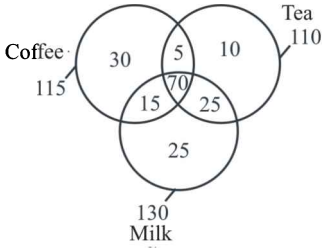
Type - 9

1. In a survey of 170 families, 115 take coffee, 110 take tea and 130 take milk. In addition, 85 coffee and milk, 75 coffee and tea, 95 tea and milk and 70 take all three. How many families take coffee and milk but not tea?

(a) 15 (b) 18
(c) 20 (d) 25

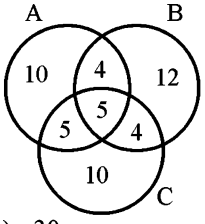
RRB RPF Constable – 19/01/2019 (Shift-I)

Ans : (a)



The number of families who take coffee and milk but not tea = 15

2. Following venn diagram represents 50 students who under go vocational training A, B & C. Each student participates in at least one training session. How many students actually participate in at least one such training session?

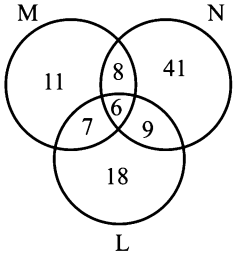


(a) 30 (b) 14
(c) 20 (d) 32

RRB RPF SI – 10/01/2019 (Shift-I)

Ans : (d) From venn diagram number of students who participate in at least one session of training session = 10 + 12 + 10 = 32

3.



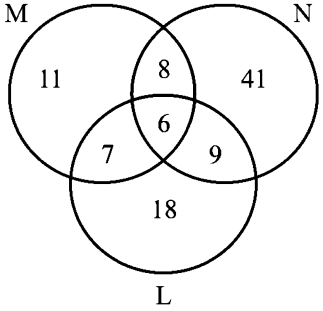
The above venn diagram shows the student who play 3 sport M, N and L. Number of students, playing sport N.

(a) 64 (b) 41
(c) 9 (d) 23

RRB RPF Constable – 25/01/2019 (Shift-I)

Ans : (a)
Total number of students who play game N
= 41 + 8 + 6 + 9
= 64

4. The venn diagram shows the three games L, M and N is played by the students in a class. Number of total students who played only one game.

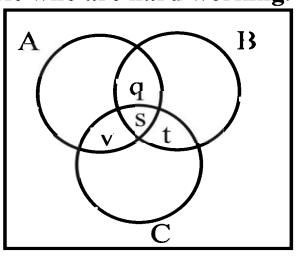


(a) 71 (b) 75
(c) 70 (d) 72

RRB Group-D – 05/10/2018 (Shift-II)

Ans. (c)
Total number of students who played only one game
= 11 + 41 + 18 = 70

5. Answer the following question using the venn diagram
A- People who are skilled.
B- People who are honest
C- People who are hard working.

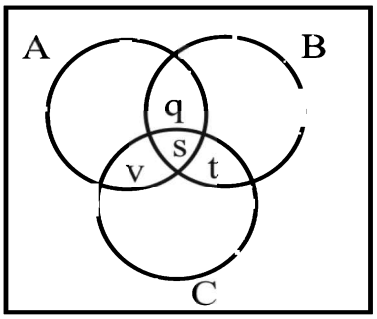


Which segment represent people who are only hard working but not honest or skilled.

(a) c-(v+t) (b) c
(c) c-(v+s+t) (d) c-(v+s)

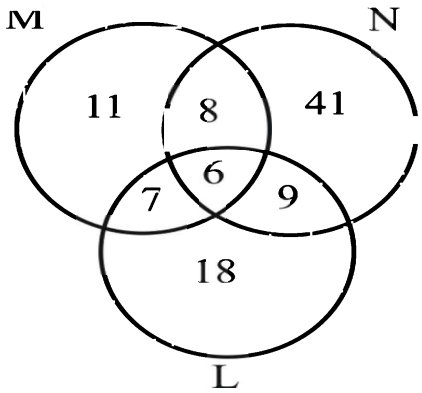
RRB Paramedical Exam – 20/07/2018 (Shift-III)

Ans. (c) :



It is clear from the Venn diagram that one who is only hardworking but not honest or skilled = c–(v+s+t)

6.



Above venn diagram shows those students who play 3 games M, N and L in the class. Number of students who play game L?

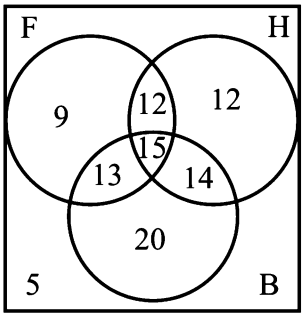
(a) 41 (b) 18

(c) 40 (d) 23

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (c) : It is clear from the given Venn diagram that the number of students, who play game L = 18+7+6+9= 40

7.



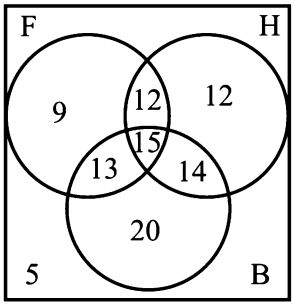
The given venn diagram shows those students who play football (F), Hockey (H) and Basket Ball (B) in the class. Total number of those students who play bastketball in class?

(a) 20 (b) 62

(c) 49 (d) 53

RRB Group-D – 24/10/2018 (Shift-I)

Ans : (b)

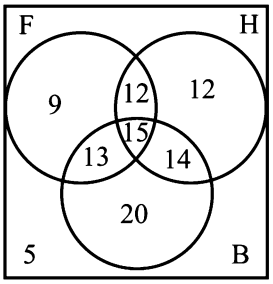


Number of students who play baseket ball

= 20+14+13+15

= 62

8. Following diagram shows no. of students play football (F), Hockey (H) and Basket Ball (B). What is the total no. of students who play the game F.



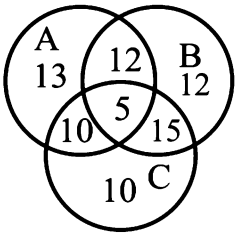
(a) 9 (b) 49

(c) 53 (d) 62

RRB Group-D – 26/10/2018 (Shift-III)

Ans : (b) According to the diagram, the number of students who play game F = 9 + 12 + 15 + 13 = 49

9. The given venn diagram represents the sports competition of a group of 77 college students. Three sports are A–Badminton, B–Basketball and C–Tennis. How many students who like to play basketball but not Tennis?



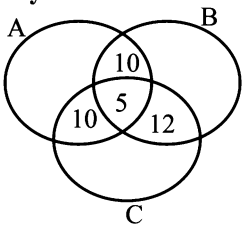
(a) 17 (b) 24

(c) 35 (d) 15

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (b) : Number of students who like to play basketball but not Tennis = 12 + 12 = 24

10. The given venn diagram shows the total 50 student. Who are sitting in three different exam A, B and C. All these who appeared in at least one examination, in how many students apeare in any one examination?

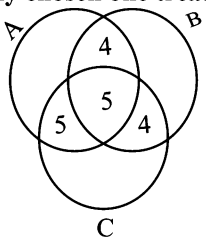


(a) 32 (b) 13
(c) 20 (d) 18

RRB Group-D – 26/11/2018 (Shift-III)

Ans : (b) Number of students who appeared in only one exam = 50 – (10 + 5 + 12 + 10)
= 13

11. The given venn diagram shows 50 patient. Those who have selected at least one of the three treatment A, B and C. How many people have actually chosen one treatment?

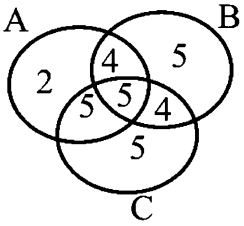


(a) 16 (b) 32
(c) 28 (d) 30

RRB Group-D – 16/11/2018 (Shift-III)

Ans. (b) : The number of patients
Who choose one treatment
= 50 – (4 + 5 + 5 + 4)
= 50 – 18
= 32

12. Given venn diagram shows total 30 shops. Which buy three different brands of biscuits (A, B and C) from a vender. Every store has purchases at least one brand from the seller. How many shops purchased brand A biscuits from the seller.

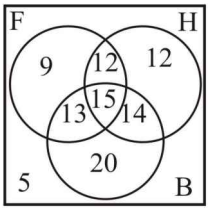


(a) 16 (b) 20
(c) 18 (d) 30

RRB Group-D – 31/10/2018 (Shift-III)

Ans : (a) Required number of persons
= 2 + 4 + 5 + 5 = 16

13.



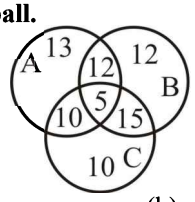
The diagram shows the number of students, playing football (F), Hockey (H) and Basketball (B) in a class. Total no. of students playing game H.

(a) 62 (b) 53
(c) 12 (d) 49

RRB Group-D – 26/10/2018 (Shift-II)

Ans : (b) Total number of students who play Hockey (H)
= 12+15+12+14
= 38+15 = 53

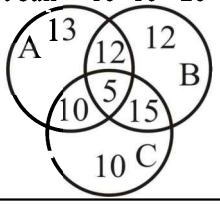
14. The given venn diagram represents the sports performances of a group of college students. Three sports A-Badminton, B-Basketball and C-Tennis. How many students play tennis but not Basketball.



(a) 20 (b) 12
(c) 17 (d) 40

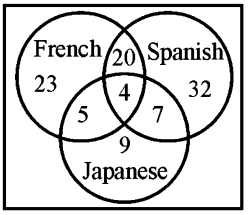
RRB Group-D – 26/10/2018 (Shift-II)

Ans : (a) Number of students who playing tennis but not basket ball = 10+10 =20



A = badminton
B = basketball
C= Tennis

15.



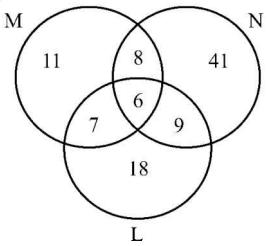
The above figure shows the number of students who study 3 languages in a class. What is the number of student studing only one language?

(a) 100 (b) 45
(c) 4 (d) 64

RRB Group-D – 25/10/2018 (Shift-II)

Ans : (d) From the given data:-
The number of students who study only one language.
= 23 + 32 + 9 = 64

16. The appropriate venn diagram shows the students who play 3 games M, N and L in a class what is the total number of students playing game 'M'.



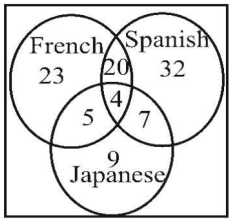
(a) 11 (b) 7
(c) 32 (d) 21

RRB Group-D – 04/10/2018 (Shift-I)

Ans. (c) Total number of students who playing game M is = 11 + 8 + 7 + 6 = 32

Type - 9

17.



The given diagram shows the number of students studying 3 lanuages in a class, what is the number of students studying only 2 languages?

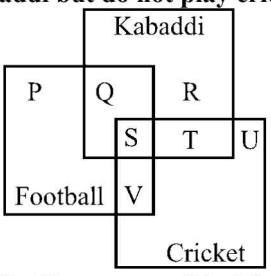
(a) 32 (b) 100
(c) 45 (d) 64

RRB RPF Constable – 17/01/2019 (Shift-III)

Ans. (a) From the given venn diagram the number of students who study only two languages

= 20 + 5 + 7 = 32

18. The students of a school play three different games. As given below, the diagram shows the number of students playing each sport, which part represents the players who play football and Kabaddi but do not play cricket.

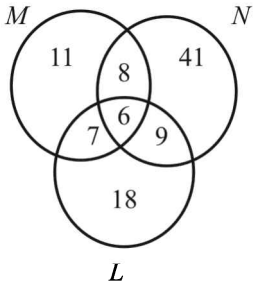


(a) P + Q + R (b) V + T
(c) S + T + V (d) Q

RRB RPF SI – 11/01/2019 (Shift-II)

Ans. (d) 'Q' plays both football and kabaddi but not plays cricket.

19.



The following venn diagram shows the students who pay 3 game M, N and L in a class, what is the total number of such students who play minimum two games?

(a) 28 (b) 31
(c) 35 (d) 30

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (d) According to the Venn diagram-

The number of students who play 3 games M, N and L in the class = 6

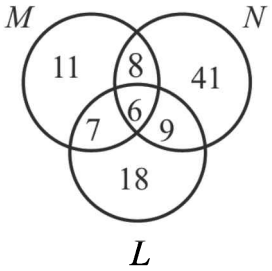
The number of students who play minimum two games

= 7 + 8 + 9 = 24

So the total students who play minimum two games

= 24+6 = 30

20.



The following venn diagram shows the students who play three games M, N and L in a class. What is the total numbers of students who are playing games N and L but not M is-

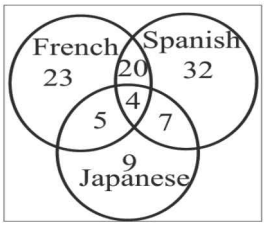
(a) 15 (b) 68
(c) 9 (d) 12

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (c) From the given Venn diagram-

Total number of students who play N and L but not play game M is '9'.

21.

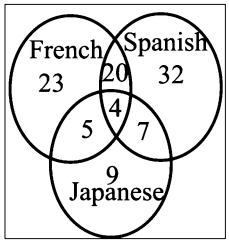


In this diagram the number of students studying three Language is shown, the number of students studying two language is-

(a) 45 (b) 100
(c) 64 (d) 32

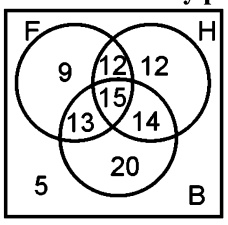
RRB Group-D – 16/10/2018 (Shift-I)

Ans. (d) :



The number of students who studying of two language
= 20+5+7
= 32

22. The diagram shows the number of students playing Football (F), Hockey (H) and Basketball (B) in a class. The total number of students actually playing two games.

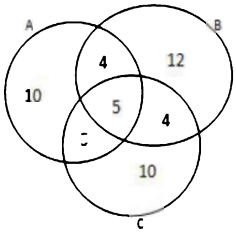


(a) 41 (b) 39
(c) 80 (d) 54

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (b) Total number of students who play two games
= 12 + 13 + 14 = 39

23. The following venn diagram shows a total of 50 students taking vocational training A, B and C. Each of which participates in at least one training session, how many students attend only two training session?

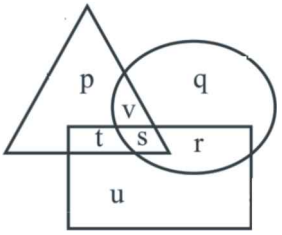


(a) 13 (b) 20
(c) 18 (d) 32

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (a) Accoding to the question-
Number of students who attend only two training sessions = 5 + 4 + 4 = 13

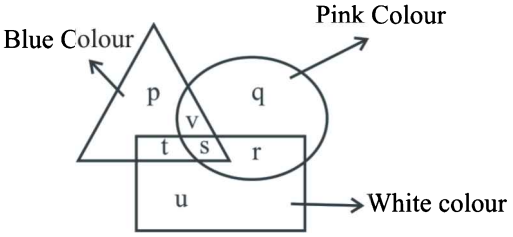
24. The following figure shows the color perference of a group of girl. The circle represents pink, the traingle represents blue and the rectangle represents white. Which group of girls like blue and pink but not white?



(a) s (b) p + v + q
(c) v (d) q + v

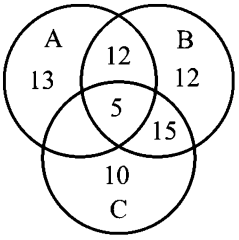
RRB Group-D – 24/10/2018 (Shift-II)

Ans. (b) :



It is clear from the figure that only P and q indicate blue colour and Pink colour whereas V indicate both blue and pink.
So Indicating blue and pink colour whereas not to white = p + v + q

25. In the given venn diagram, the game preference for a group of 77 college students is given. The three games are given as A, B and C. How many students like to play C and A but not B.

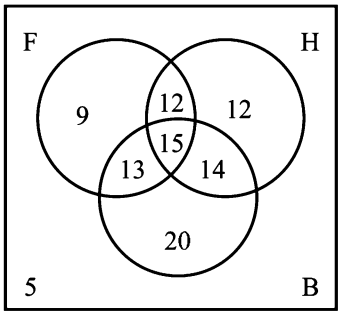


(a) 15 (b) 33
(c) 10 (d) 23

RRB Group-D – 26/10/2018 (Shift-III)

Ans : (d)
∵Accoding to the venn diagram the number of students who like game C and A but not like game B.
= 10+13 = 23

26.



The given diagram shows the number of students playing game football (F), Hockey (H) and Basketball (B) in the class. The total number of students playing maximum two games is.

(a) 39 (b) 54
(c) 41 (d) 80

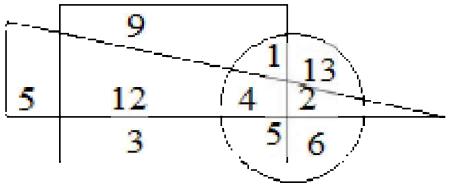
RRB Group-D – 31/10/2018 (Shift-I)

Ans : (d) Total number of students who like to play maximum two games.

= liking one game + liking two game

9 + 12 + 20 + 12 + 13 + 14 = 80

27. In the following venn diagram the rectangle represents the number of RPS supporters, the traingle represents the number of MI supporters and the circle represents the number of KKR supporters. Find the numbers of supporters who support the KKR and only one other team.

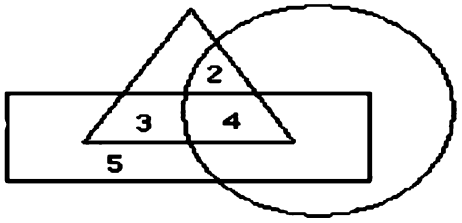


(a) 8 (b) 10
(c) 2 (d) 6

RRB Group-D – 07/12/2018 (Shift-III)

Ans : (a) Numbers of team who support KKR and only another team 5 + 1 + 2 = 8

28. In the given venn diagram, the triangle represents the healthy people, rectangle represents old people and the circle represents men. What is the number of healthy but old people?

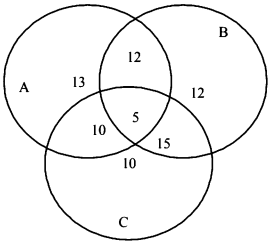


(a) 4 (b) 3
(c) 2 (d) 6

RRB Group-D – 28/11/2018 (Shift-I)

Ans : (a) From the given venn diogram it is clear that the number of healthy but old people = 4

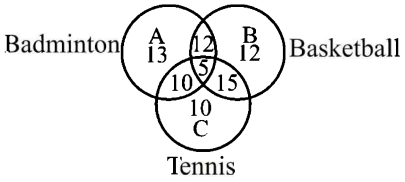
29. The given venn diagram represents the sports preferences of a group of college students. Badminton for A, B for Basketball and C for Tennis. How many students actually like to play two of these three games.



(a) 35 (b) 17
(c) 5 (d) 37

RRB Group-D – 31/10/2018 (Shift-I)

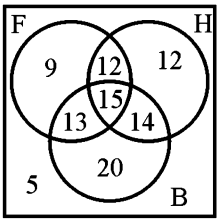
Ans. (d) :



The number of students who like two of these three games

= 12+15+10=37

30. The given diagram shows the number of students playing football (F), Hockey (H) and Basketball (B).



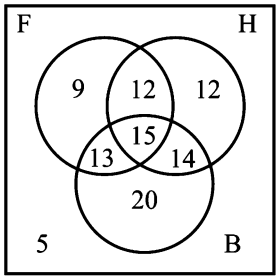
What is the number of students playing at least two games?

(a) 41 (b) 39
(c) 54 (d) 80

RRB Group-D – 15/11/2018 (Shift-II)

Ans : (c) The number of students who play at least two games = 12 + 13 + 15 + 14 = 54

31. The given venn diagram shows the number of students playing football (F), Hockey (H) and Bastketball (B) in a class.



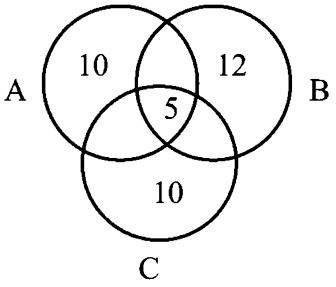
Find the total number of student playinig F and H but not B.

(a) 33 (b) 21
(c) 12 (d) 27

RRB Group-D – 12/11/2018 (Shift-III)

Ans : (c) The number of students who playing F and H but not playing B is = 12

32. The following venn diagram shows the 50 students who attend the vocational training session A, B and C. Each student participates in at least one training session. How many students actually participate in only 2 sessions in the training?

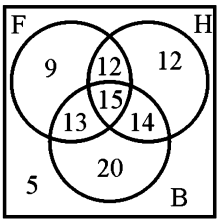


(a) 8 (b) 4
(c) 13 (d) 5

RRB Group-D – 12/11/2018 (Shift-I)

Ans. (c) : The number of students who participate only in two sessions = Number of total students – (number of students who Participating one or more than two sessions)
= 50 – (12 + 10 + 10 + 5)
= 50 – 37 = 13

33. The following diagrams shows the number of students playing football (F), Hockey (H) and Basketball (B) in the class, what is the total number of students playing F and B but not playing H?



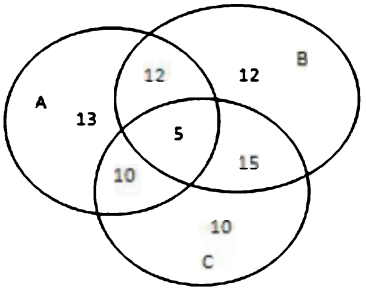
(a) 28 (b) 29
(c) 42 (d) 13

RRB Group-D – 05/11/2018 (Shift-II)

Ans : (d) From the venn diagram, the number of students who play F and B but not play H is 13.

34. The given venn diagram represents the sports preferences of a group of college students. Three sports are Batminton-A, Basketball-B and Tennis-C. How many students play badminton and basketball but not tennis?

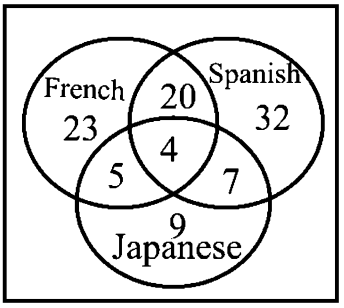
(a) 37 (b) 15
(c) 30 (d) 20



RRB Group-D – 01/11/2018 (Shift-II)

Ans : (a) From the given Venn diagram
Number of players who play Badminton or Basketball play but not Tennis = 13+12+12 = 37

35. Answer the following question by consider the venn diagram below.

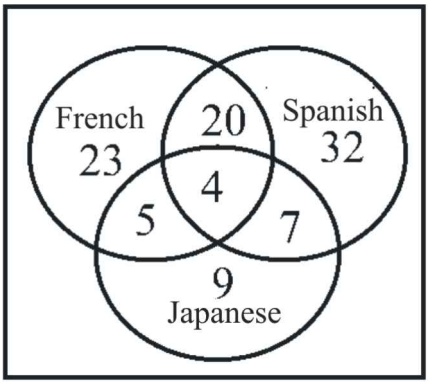


The venn diagram shows students studying 3 subjects in a class. The number of students studying French and Spanish but not studying Japanese is:

(a) 29 (b) 4
(c) 20 (d) 24

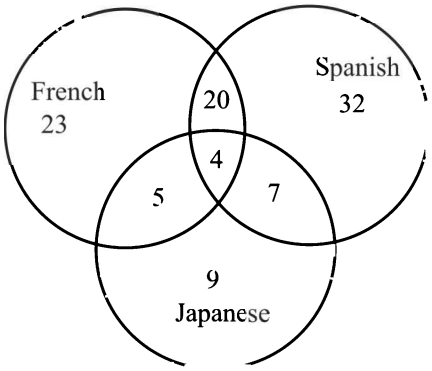
RRB Group-D – 24/10/2018 (Shift-III)

Ans. (c) :



From the following venn diagram it is clear that the number of students who read French and Spanish but not Japanese = 20

36.



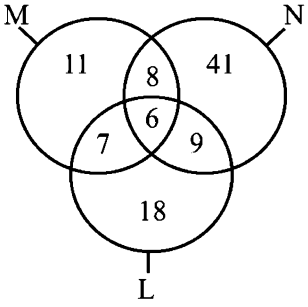
This diagram shows the number of students studying three languages in the classroom. How many students who study Spanish and Japanese but not frence.

(a) 7 (b) 11
(c) 32 (d) 4

RRB Group-D – 23/10/2018 (Shift-III)

Ans : (a) The number of students who study Spanish and Japanses but not French = 7

37.



The following venn diagram shows the three games played, M, N and L in a class. Find the total number of students who played M and L but not play N.

(a) 36 (b) 13
(c) 15 (d) 7

RRB Group-D – 05/10/2018 (Shift-III)

Ans. (d) It is clear from the venn diagram that-
total number of students playing game M and L but not playing game N = 7

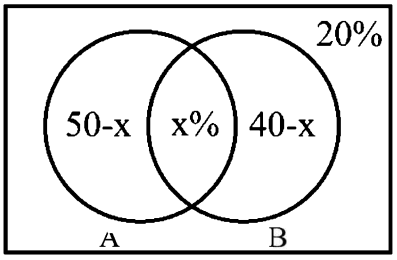
Type - 3

50% of people read 'A' newspaper, 40% of people read 'B' newspaper, 20% of people read neither newspaper A nor newspaper B. If the number of people who read both newspapers is 500. Then how many people were surveyed.

(a) 7000 (b) 4500
(c) 5000 (d) 3000

RRB RPF SI – 13/01/2019 (Shift-III)

Ans : (c)



from the venn diagram,

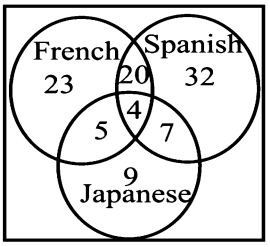
$50 - x + x + 40 - x + 20 = 100$

$110 - x = 100$

$x = 10\%$

10% = 500
100% = 5000
So Total number of people is 5000 in the survey.

39.



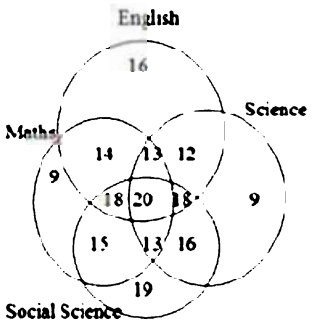
The following figure shows the number of students who study three languages in the class. Total number of students studying three languages French, Spanish and Japanese is.

(a) 100 (b) 4
(c) 16 (d) 41

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (b) It is clear from the figure that the number of students is 4 who studying all the three language.

40. Study the following venn diagram are carefully and answer the question given below.



What is the total number of students who have taken English or Mathematics or social science?

(a) 185 (b) 180
(c) 183 (d) 175

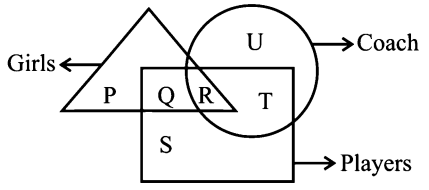
RRB Group-D – 07/12/2018 (Shift-III)

Ans : (c) From the given diagram, total number of students of english, maths and social science

$\Rightarrow 16 + 14 + 9 + 18 + 15 + 19 + 13 + 12 + 20 + 18 + 13 + 16 = 183$

So the total number of students of all section is 183 except science students.

41. In the following figure the triangle represents the girls and the square represents players and circle represents coach. Which part of the figure represents girls who are players and coach also?

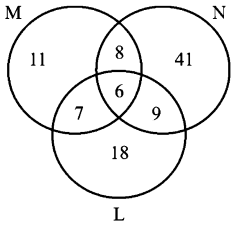


(a) R (b) S
(c) Q (d) P

RRB Group-D – 26/11/2018 (Shift-III)

Ans : (a) From the given figure it is clear that diagram's "R" part represents those girls who are player as well as coach.

42. The given venn diagram represents the students participating from a class in three types of games M, N and L.



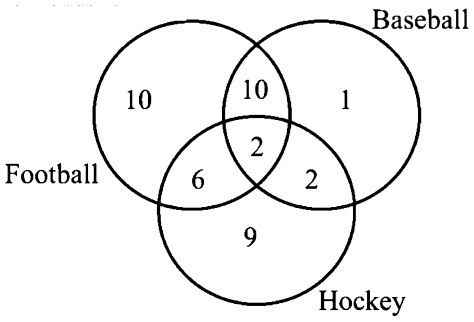
(a) 64 (b) 6
(c) 100 (d) 30

RRB Paramedical Exam – 21/07/2018 (Shift-I)

And. (b) From the following venn diagram it is clear that total number of students who participating in three games is 6.

43. The given venn diagram shows the students who like Football, Basketball and Hockey in the class of 40 students. The total number of students who like at least one sport is.

Total students = 40



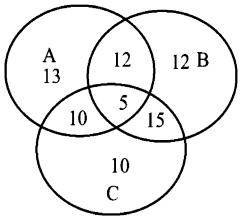
(a) 40 (b) 30
(c) 10 (d) 20

RRB Group-D – 16/11/2018 (Shift-I)

Ans. (a) Hence sum of students is 40 in given Venn diagram.

So all students like minimum one game.

44. The given venn diagram represents the game priorities of the group of college students. Three games are A-Badminton, B-Basketball and C-Tennis. How many students play at least one game?



(a) 77 (b) 35
(c) 17 (d) 5

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (a) :

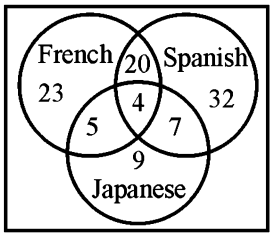
Number of students playing only one game = 13 + 12 + 10 = 35

Students playing at least two games = 12 + 10 + 15 = 37

Number of students playing all three games = 5

So number of students who playing minimum one game is = 35 + 37 +5 = 77

45.



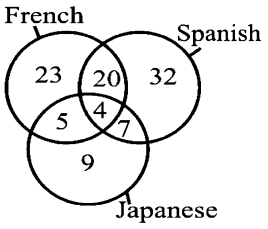
The venn diagram shows the number of students studying three languages in a class. What is the total number of students studying in the class?

(a) 100 (b) 45

(c) 101 (d) 64

RRB Group-D – 23/10/2018 (Shift-I)

Ans. (a) :

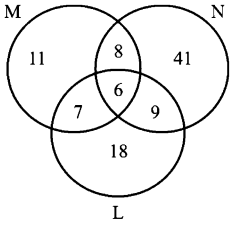


Total number of students who studying in the class.

= 23+20+32+5+4+7+9

= 100

46.



The above venn diagram shows the students who are playing three games M, N and L in a class. What is the total number of students in the class?

(a) 100 (b) 102

(c) 101 (d) 98

RRB Group-D – 08/10/2018 (Shift-I)

Ans. (a) : Total number of students in M, N and L class = 11 + 8 + 41 + 6 +7 + 9 + 18 = 100

Type - 4

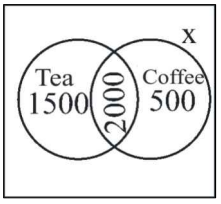
47. The number of students in a college is 5000. 3500 students like coffee, 2500 students like tea and 2000 students like both tea and coffee. How many students don't like either of these two drinks?

(a) 1500 (b) 1000

(c) 500 (d) 2000

RRB RPF Constable – 22/01/2019 (Shift-III)

Ans. (b) Let the number of students is X who do not like both



5000 = 2000 + 1500 + 500 + x

x = 5000 – 4000

x = 1000

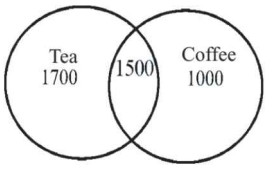
48. In a town with a population of 5000, 3200 people take tea, 2500 people take coffee and 1500 people take both tea and coffee. How many of them neither take tea nor coffee?

(a) 800 (b) 770

(c) 900 (d) 1800

RRB Group-D – 15/10/2018 (Shift-I)

Ans : (a)



The number of people who take only tea

= 3200 – 1500 = 1700

The number of people who take only coffee

= 2500 – 1500 = 1000

Number of people Neither take tea nor coffee

= 5000 – (1700 + 1000 + 1500)

= 5000 – 4200 = 800

So, 800 people neither take tea nor coffee.